You are building the mobile personnel reporting flow for "Raynham Estate OS" — a premium Progressive Web App for a rural English country estate. This build covers EXACTLY three screens plus the identity-claim sheet and their states. Do not create navigation bars, tab bars, dashboards, login screens, settings, or any other screens. If a requirement is ambiguous, choose the simplest option and note your assumption in code comments.

1. WHAT YOU ARE BUILDING

A QR-scan-triggered, login-free reporting flow for field staff, tenants, and contractors. A user scans a physical QR tag (URL pattern /a/{anchorId}) and lands on a three-choice panel. From there they can view information, report an issue with a photo, or request emergency help. Staff may OPTIONALLY claim their identity once per device; identification improves attribution and safety but must NEVER be required to report or to get help.

Hard product rules:
1. No login, signup, password, or app download at any point in this flow. The identity-claim sheet is a convenience for staff, never a gate: every action in this build is fully usable without it.
2. The full report flow must complete in under 30 seconds with no typing required (photo + one category chip + submit).
3. Every screen must function on a throttled 3G connection and degrade gracefully offline.
4. All touch targets are a minimum of 44px. Design for gloved hands, bright sunlight, and older users.
5. The colour red appears ONLY on the emergency entry points. Any other use of red is a defect.
6. Plain language on all user-facing text. No feature names, no jargon, no "AI" branding.

2. TECH STACK

- Next.js (App Router) + React + TypeScript
- Tailwind CSS, with the design tokens below mapped to CSS custom properties on :root — components must consume token classes/variables, never raw hex
- next-pwa (or equivalent service worker) for installability and offline cache
- IndexedDB (via idb-keyval or Dexie) for the offline report outbox
- localStorage for the claimed staff profile (key: "raynham.staffProfile")
- Mock API layer: implement the data contracts in section 5 as a local mock

(in-memory + IndexedDB persistence). Keep every call behind a small /lib/api module so a real backend can replace it without touching UI code.

3. DESIGN TOKENS (LIGHT MODE — INLINE SOURCE OF TRUTH)

Apply these to :root exactly:

```
:root {
  --surface-base:    #FAFAF7;
  --surface-raised:  #FFFFFF;
  --surface-sunken:  #F3F2EC;
  --border-subtle:   #E6E4DB;
  --border-strong:   #D4D1C4;
  --text-primary:    #2E3438;
  --text-secondary:  #5B6770;
  --text-muted:      #8A9299;
  --brand-gold:      #C9A227;
  --brand-gold-hover:#B08D1E;
  --brand-gold-soft: #F5EBCE;
  --status-success:  #2F7D4F;
  --status-success-soft: #E3F0E7;
  --status-warning:  #C9821E;
  --status-warning-soft: #F9EEDC;
  --status-danger:   #C43D2E;  /* EMERGENCY ENTRY POINTS ONLY */
  --status-danger-soft:  #F8E3DF;
  --status-neutral:  #6B7280;
  --focus-ring:      rgba(201, 162, 39, 0.45);
}
```

Typography: Inter (system-ui fallback). Mono (What3Words + reference codes): "IBM Plex Mono", ui-monospace, monospace.
Radii: 6px / 10px / 16px / pill. Shadows: soft, low-opacity slate shadows only (e.g. 0 1px 2px rgba(46,52,56,0.06), 0 4px 12px rgba(46,52,56,0.06)).
Spacing scale: 4 / 8 / 12 / 16 / 20 / 24 / 32px.

4. THE THREE SCREENS (+ IDENTITY SHEET)

Match the attached mockups for layout, hierarchy, and tone. Where the mockups and this spec disagree, this spec wins for behaviour and the mockups win for visual composition.

SCREEN A — Scan Landing (route: /a/[anchorId])

1. Slim white header: small monogram logo (a stylised letter "R", CSS gradient from --brand-gold at top to #5B6770 at bottom) above a tiny letter-spaced uppercase wordmark "RAYNHAM"; right side shows a small GSM signal icon.
2. Identity card (white, --border-subtle border, 10px radius): bold title "{assetName} — {zoneName}" (e.g. "Tractor 3 — Home Farm"); beneath it a monospace chip "///{w3w}" with a copy icon that copies the address and shows a 2s "Copied" toast.
3. Active Notice banner — render ONLY if anchor.activeNotice exists and is not expired: soft amber background (--status-warning-soft), 4px amber left border, plain-language body text (e.g. "Martin is due to check this machinery at 4 p.m. Please do not use until verified."), with the expiry time in muted text. This banner must appear ABOVE the three choice cards and must not be dismissible by the scanner.
4. Three full-width stacked cards, each 72px tall, white surface, soft shadow, 10px radius, 4px coloured left accent bar, leading icon, title in semibold slate, muted one-line subtitle:
   - "Information & Updates" (green accent, subtitle "Site notices & guidance")
     → routes to a simple read-only notices list for this anchor/zone.
   - "Report a Concern" (amber accent, subtitle "Photo report, 30 seconds")
     → Screen B.
   - "Get Help Now" (red accent, subtitle "Alerts the on-call manager")
     → for this build, a stub confirm sheet reading "This would alert the on-call manager with your location." Mark with a TODO for the safety build.
5. Footer: centred muted micro text "No app or login needed".
6. States: loading skeleton while the anchor resolves; a full-width offline banner ("No signal — showing saved information") when navigator.onLine is false, served from the service-worker cache.
7. IDENTITY (per the Identity & Onboarding Model):
   a. If NO staff profile exists in localStorage: render a subtle text button directly beneath the identity card — "Site staff? Identify yourself" in --text-secondary. Tapping opens the Identity Sheet (below). This link is quiet and secondary; it must not compete with the three choice cards.
   b. If a profile EXISTS: render a small muted line in the same position — "Signed in as {name}" with a "Not you?" text button that clears the stored profile and reopens the Identity Sheet.
   c. The lone-worker entry point (the stub in this build) also triggers the Identity Sheet on first use if no profile exists — framed as "We need to know who to check on."

IDENTITY SHEET (bottom sheet, mobile):

1. Title "Who are you?" with muted subtext "One quick step — so the team knows who's on site."
2. A name field with type-ahead lookup across the seeded staff list AND any previously self-registered profiles (search after 2 characters, 200ms debounce). If the typed name matches nothing, offer "Continue as {name}" to self-register.
3. A mobile number field (tel input, UK format accepted, store as E.164).
4. If the config flag SMS_VERIFICATION_ENABLED is true: after confirm, show a 6-digit code step (mock the code as 123456 in this build) before saving. If false, skip straight to saving.
5. Primary button "Confirm" (full-width, --brand-gold, 56px); secondary text button "Not now" that dismisses the sheet. Dismissing never blocks anything — the user simply continues anonymously.
6. On save: persist StaffProfile to localStorage, show a 2s success toast "Thanks {name} — you're set", close the sheet, and update Screen A's identity line.
7. Duplicate-guard: if the entered mobile already exists on another profile, merge/confirm by name match rather than creating a second record.

SCREEN B — Report a Concern (route: /a/[anchorId]/report)

1. Header bar: back chevron (returns to Screen A), centred title "Report a Concern", small amber dot as the section marker.
2. Camera-first capture card: on mount, request the device camera (getUserMedia, facingMode "environment"); live viewfinder with a large round capture button. After capture, show the photo thumbnail with a "Retake" text button. If camera permission is denied, fall back to a native file-input picker — never a dead end. Compress client-side (max 1600px longest edge, JPEG ~0.7 quality).
3. Label "What's the issue?" above a wrapping grid of category chips (10px radius, white, --border-strong border): Machinery · Fence & Gates · Livestock · Tree / Forestry · Water & Pumps · Building · Grounds · Other. Single-select; selected chip uses --status-warning-soft fill, amber border, dark text. Exactly one chip required to enable submit.
4. Note field: single-line input, placeholder "Add a note (optional)", with a microphone icon for voice-to-text (Web Speech API where supported; hide otherwise).
5. Auto-capture location row (muted, small): pin icon + "Location captured — ///{w3w}" in mono with a checkmark. Attempt navigator.geolocation (5s timeout); on denial or timeout silently use the anchor's stored location — scanning the physical tag already proves presence. Never block submission on location.
6. ATTRIBUTION: if a StaffProfile exists in localStorage, include

reportedBy: { name, mobile } in the submission payload. If none exists, submit with reportedBy: null — do NOT prompt for identity here (the report must stay at three taps).
7. Submit block: above the button, a small muted line "Sends to the right team automatically." Then one full-width primary button, 56px tall, --brand-gold fill, dark slate text, label "Submit Report". Disabled until photo + one chip are present. Spinner state while sending.
8. Submission behaviour: route via CATEGORY_ROUTING (section 5), persist, navigate to Screen C. Offline: save to the IndexedDB outbox, navigate anyway (optimistic), toast "No signal — your report will send automatically when reconnected." Flush the outbox automatically on the online event.

SCREEN C — Confirmation (route: /a/[anchorId]/report/sent)

1. Upper third: large soft-green circle with a white checkmark; bold heading "Report received"; muted subheading "You can get back to work."
2. "SENT TO" card (white, thin border): micro uppercase muted label "SENT TO", then "{workspace} — {team}" in slate semibold (e.g. "Property & Grounds — Maintenance Team"), an amber "Assigned" status chip, relative timestamp "just now". If reportedBy exists, add a small muted line "Reported by {name}".
3. Reference card (white, thin border), two mono rows each with a copy icon: "Ref #RN-0427" (generate sequential-style IDs) and "Location ///{w3w}".
4. Muted three-step mini list titled "What happens next": "1. The team is notified" · "2. You'll see updates if you rescan the tag" · "3. The tag shows when it's fixed".
5. If the report is still queued offline, show a small neutral banner: "Queued — will send when reconnected."
6. Bottom: full-width white secondary button (grey border) "Done" → returns to Screen A. Beneath it, a small red-tinted text link "Still need urgent help? Get Help Now" → the same stub sheet as Screen A.

5. DATA CONTRACTS & ROUTING

```
interface Anchor {
  id: string;
  assetName: string;            // "Tractor 3"
  zoneName: string;             // "Home Farm"
  workspace: 'farm' | 'property' | 'hospitality';
  lat: number; lng: number;
  w3w: string;                  // "filled.count.soap" (no leading slashes)
```

```
  activeNotice?: { body: string; expiresAt: string; author: string } | null;
}

interface StaffProfile {
  id: string;
  name: string;
  mobile: string;                // E.164
  workspace: 'farm' | 'property' | 'hospitality';
  verified: boolean;             // true once SMS code confirmed (if enabled)
  enrolledBy: 'manager' | 'self';
  deviceBoundAt: string;
}

type Category = 'Machinery' | 'Fence & Gates' | 'Livestock' | 'Tree / Forestry'
  | 'Water & Pumps' | 'Building' | 'Grounds' | 'Other';

interface Report {
  id: string;                    // "RN-0427"
  anchorId: string | null;       // null allowed for future pin-drop reports
  category: Category;
  note?: string;
  photoUrl: string;              // blob URL in the mock layer
  lat: number; lng: number; w3w: string;
  reportedBy: { name: string; mobile: string } | null;
  routedTo: { workspace: string; team: string };
  status: 'new' | 'assigned' | 'in_progress' | 'resolved';
  createdAt: string;
}
```

CATEGORY_ROUTING (static config map — PLACEHOLDER for the future server-side classification step; keep in one exported config file with a comment saying so):
- Machinery, Fence & Gates, Livestock, Tree / Forestry, Water & Pumps
  → { workspace: 'Farm & Land', team: 'Farm Team' }
- Building, Grounds
  → { workspace: 'Property & Grounds', team: 'Maintenance Team' }
- Other
  → { workspace: 'Unassigned', team: 'Estate Office' }

Seed the mock layer with:
- Three anchors: "Tractor 3 — Home Farm" (WITH the Active Notice from the mockup), "Yurt 4 — Glamping Meadow", and "12 Estate Cottages — Residential" (no notice).
- A pre-loaded staff list of five profiles (enrolledBy: 'manager'), including "Martin Clarke" (farm) and "Dan Reeves" (property) so the type-ahead lookup

has real data.
- Config flag SMS_VERIFICATION_ENABLED = false by default.

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6. ACCESSIBILITY & QUALITY BAR
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1. WCAG 2.2 AA contrast throughout; status never conveyed by colour alone (icons and labels accompany every colour).
2. All interactive elements ≥ 44px; flows fully operable one-handed.
3. Respect prefers-reduced-motion; the only animation allowed is a subtle card fade/slide on navigation.
4. Accessible names on all buttons; aria-label on the camera viewfinder.
5. First load of the scan landing under 100KB of JS excluding mock assets.

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7. ACCEPTANCE CRITERIA (VERIFY BEFORE DECLARING DONE)
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1. GATE TEST (critical): on a fresh browser with no profile, a user can reach the confirmation screen via the report flow AND trigger the Get Help Now stub — with zero identification and no prompt blocking either path.
2. First-use identity claim completes in under 60 seconds: name lookup + mobile confirm (+ mock code 123456 if SMS verification is enabled).
3. Once claimed, identity survives a full browser restart on the same device; Screen A shows "Signed in as {name}"; "Not you?" clears it.
4. A report submitted while identified carries reportedBy and Screen C shows "Reported by {name}"; an anonymous report carries reportedBy: null and renders identically otherwise.
5. Offline submission queues, lands on Screen C with the queued banner/toast, and sends automatically on reconnection.
6. Camera denial falls back to file picker; geolocation denial completes with the anchor location.
7. Red appears only on the two emergency entry points; grep the components — no other occurrence of the danger token.
8. All styles resolve to token variables; no raw hex in component files. No password fields exist anywhere.
9. Zero TypeScript errors; screens render correctly at 360px and 390px widths.